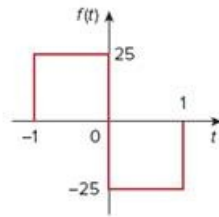


Problem

Obtain the Fourier transform of the function in Fig. 18.6.

Figure 18.6



Step-by-step solution

Step 1 of 2

Refer to the function $f(t)$ in Figure 18.6 in the textbook.

The function $f(t)$ is expressed as,

$$f(t) = \begin{cases} 25, & -1 \leq t < 0 \\ -25, & 0 < t \leq 1 \end{cases}$$

The Fourier transform of the function $f(t)$ is,

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ &= \int_{-1}^0 25 e^{-j\omega t} dt + \int_0^1 (-25) e^{-j\omega t} dt \\ &= 25 \int_{-1}^0 e^{-j\omega t} dt - 25 \int_0^1 e^{-j\omega t} dt \\ &= 25 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1}^0 - 25 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1 \end{aligned}$$

Step 2 of 2

Further simplify the transform $F(\omega)$.

$$\begin{aligned} F(\omega) &= -\frac{25}{j\omega} [e^0 - e^{\omega}] + \frac{25}{j\omega} [e^{-\omega} - e^0] \\ &= -\frac{25}{j\omega} [1 - e^{\omega}] + \frac{25}{j\omega} [e^{-\omega} - 1] \\ &= \frac{25}{j\omega} [e^{-\omega} - 1 - 1 + e^{\omega}] \\ &= \frac{25}{j\omega} [e^{\omega} + e^{-\omega} - 2] \end{aligned}$$

Since $\cos \omega = \frac{e^{\omega} + e^{-\omega}}{2}$,

$$\begin{aligned} F(\omega) &= \frac{25}{j\omega} [2 \cos \omega - 2] \\ &= \frac{50(\cos \omega - 1)}{j\omega} \end{aligned}$$

Thus, the Fourier transform of the function $f(t)$ is $\boxed{\frac{50(\cos \omega - 1)}{j\omega}}$.